INCH POUND

MS3121E
w/AMENDMENT 1
19 October 2017
SUPERSEDING
MS3121E
15 January 2015

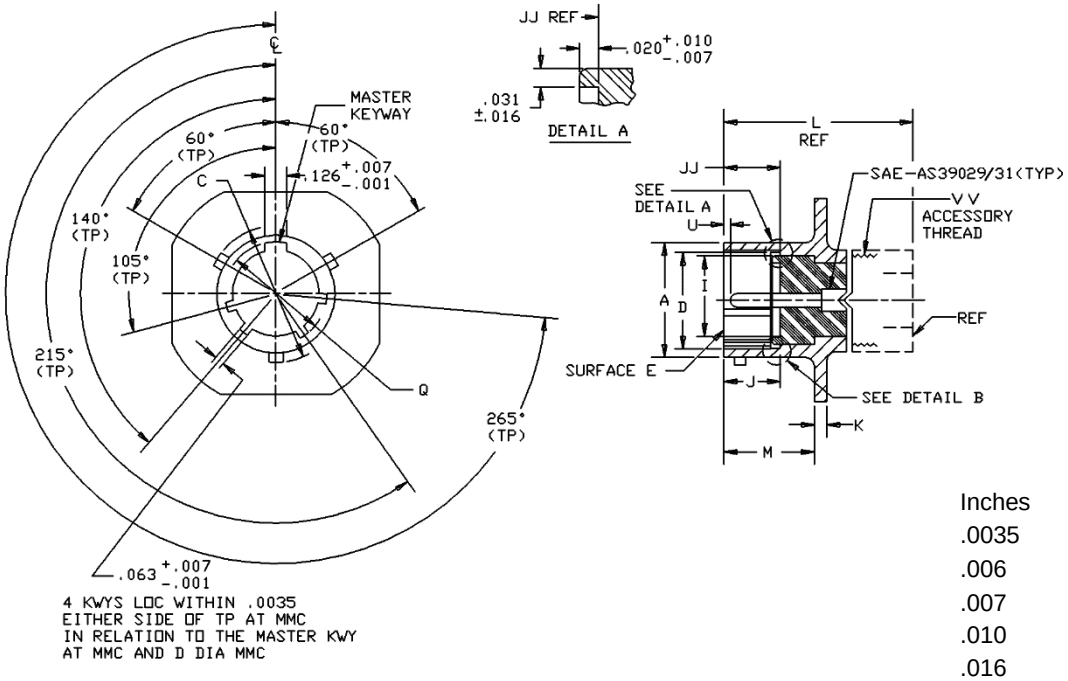
DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, CRIMP TYPE, CABLE-CONNECTING, BAYONET COUPLING, SERIES 1, CLASSES E, F AND P

Inactive for new design after 5 September 1975.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-26482.



Inches	mm
.0035	0.089
.006	0.152
.007	0.178
.010	0.254
.016	0.406
.020	0.51
.031	0.79
.063	1.60
.126	3.22

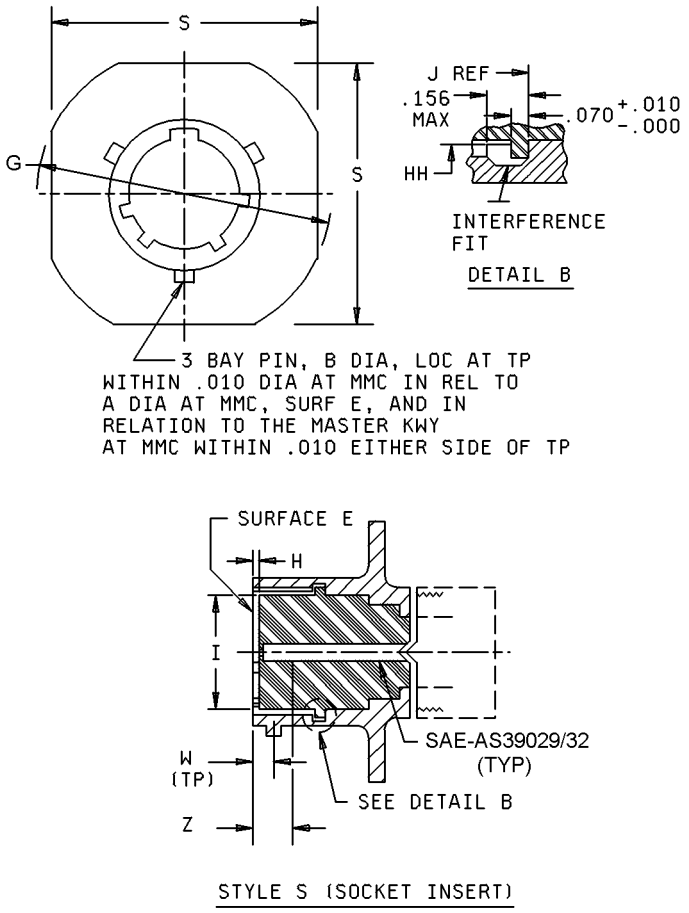
STYLE P (PIN INSERT)

FIGURE 1. Plug, dimensions, classes E, F and P.

AMSC N/A

FSC 5935

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STYLE S (SOCKET INSERT)

Inches	mm
.010	0.254
.070	1.78
.156	3.96

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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Shell size	A dia +.001 -.005 OD	B dia +.006 -.002 bay pin	C dia +.000 -.016 over bay pins	D dia +.005 -.001 ID	G dia ±.020 across flange corners	H +.000 -.020 socket insert location
8	.473 (12.01)	.078 (1.98)	.563 (14.30)	.362 (9.19)	.938 (23.83)	.025 (0.64)
10	.590 (14.99)		.680 (17.27)	.490 (12.45)	1.062 (26.97)	
12	.750 (19.05)		.859 (21.82)	.607 (15.42)	1.156 (29.36)	
14	.875 (22.23)		.984 (24.99)	.732 (18.59)	1.250 (31.75)	
16	1.000 (25.40)		1.108 (28.14)	.857 (21.77)	1.344 (34.14)	
18	1.125 (28.58)		1.233 (31.32)	.962 (24.43)	1.438 (36.53)	
20	1.250 (31.75)		1.358 (34.49)	1.087 (27.61)	1.562 (39.67)	.087 (2.21)
22	1.375 (34.93)		1.483 (37.67)	1.212 (30.78)	1.688 (42.88)	
24	1.500 (38.10)	.125 (3.18)	1.610 (40.89)	1.337 (33.96)	1.812 (46.02)	

FIGURE 1. Plug, dimensions, classes E, F and P – Continued.

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Shell size	I max insert dia	J ±.010 packing location	JJ +.000 -.020 pin insert location	K ±.016 thick mounting flange	M +.031 -.000 mounting flange location
8	.285 (7.24)	.382 (9.70)	.332 (8.43)	.094 (2.39)	.400 (10.16)
10	.402 (10.21)				
12	.516 (13.11)				
14	.641 (16.28)				
16	.766 (19.46)				
18	.855 (21.72)				
20	.980 (24.89)	.444 (11.28)	.394 (10.01)	.115 (2.92)	.535 (13.59)
22	1.105 (28.07)				
24	1.229 (31.22)				.568 (14.43)

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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Shell size	Q dia +.005 -.006 over keyways	S max length side	HH max ID gasket	U +.010 -.020 pin contact location
8	.412 (10.46)	.828 (21.03)	.329 (8.36)	.085 (2.16)
10	.540 (13.72)	.954 (24.23)	.457 (11.61)	
12	.689 (17.50)	1.047 (26.59)	.564 (14.33)	
14	.814 (20.68)	1.141 (28.98)	.689 (17.50)	
16	.939 (23.85)	1.234 (31.34)	.814 (20.68)	
18	1.039 (26.39)	1.326 (33.68)	.907 (23.04)	
20	1.164 (29.57)	1.453 (36.91)	1.039 (26.39)	.147 (3.73)
22	1.289 (32.74)	1.578 (40.08)	1.164 (29.57)	
24	1.414 (35.92)	1.703 (43.26)	1.289 (32.74)	

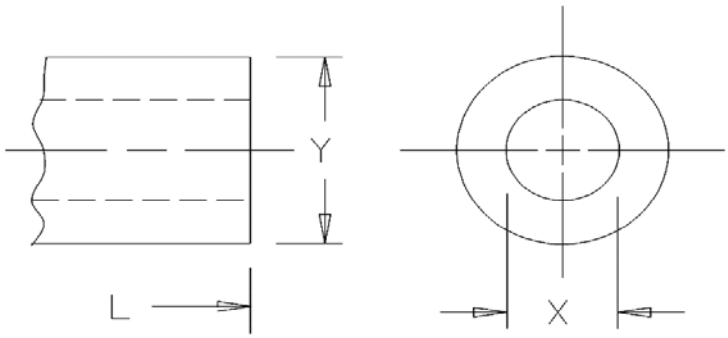
FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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Shell size	W (TP) bay pin location	Z (b) +.000 -.078 socket contact spring location	V V class 2A accessory thread
8	.100 (2.54)	.153 (3.89)	7/16-28
10			9/16-24
12			11/16-24
14			13/16-20
16			15/16-20
18			1-1/16-18
20		.215 (5.46)	1-3/16-18
22			1-5/16-18
24	.109 (2.77)		1-7/16-18

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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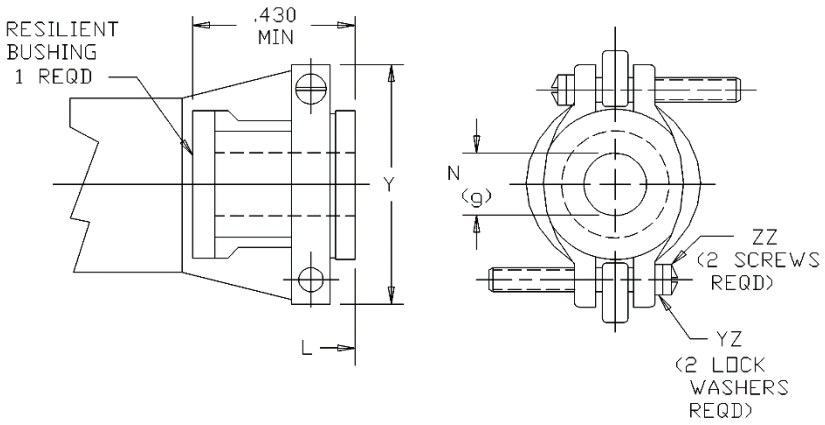
Dimensions for class E

Shell size	L max over-all length	X dia min ID	Y dia max OD
8	1.552 (39.42)	.259 (6.58)	.608 (15.44)
10		.359 (9.12)	.734 (18.64)
12		.469 (11.91)	.858 (21.79)
14		.589 (14.96)	.984 (24.99)
16		.717 (18.21)	1.110 (28.19)
18		.779 (19.79)	1.234 (31.34)
20	1.709 (43.41)	.901 (22.89)	1.360 (34.54)
22		1.009 (25.63)	1.484 (37.69)
24		1.123 (28.52)	1.610 (40.89)

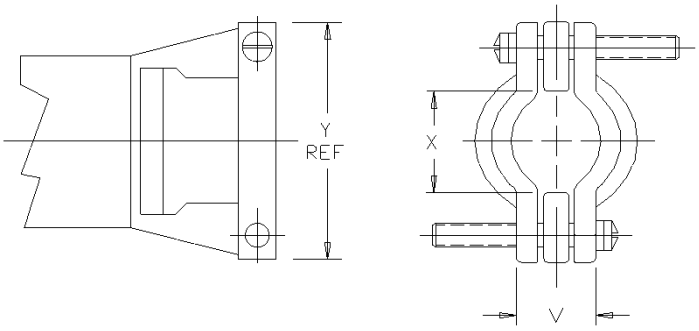
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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Dimensions for class F



Inches	mm
.430	10.92

Dimensions for class F, bushing removed

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

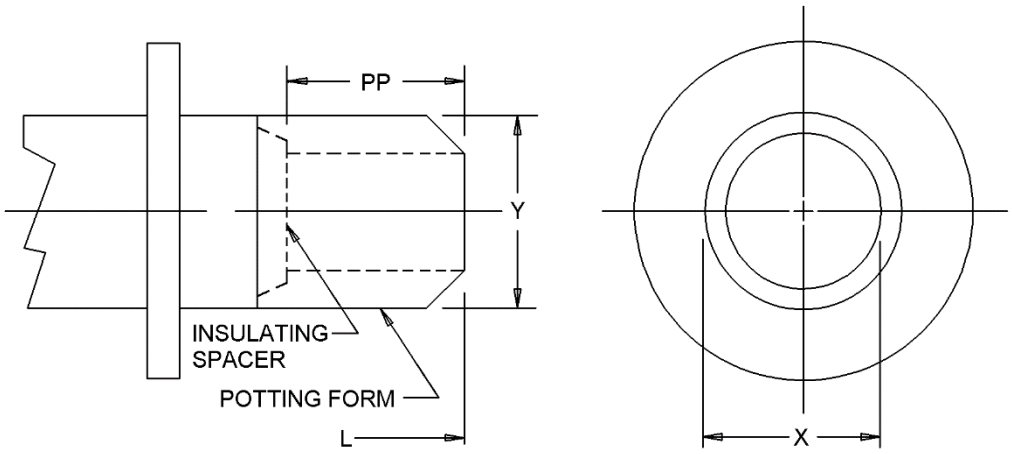
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Shell size	L max over-all length	X dla min open	Y max	V max closed
8	2.422 (61.52)	.234 (5.94)	.828 (21.03)	.187 (4.75)
10		.297 (7.54)	.891 (22.63)	.187 (4.75)
12		.422 (10.72)	1.016 (25.81)	.281 (7.14)
14		.547 (13.89)	1.141 (28.98)	.325 (8.26)
16	2.537 (64.44)	.609 (15.47)	1.203 (30.56)	.356 (9.04)
18		.734 (18.64)	1.469 (37.31)	.456 (11.58)
20	2.824 (71.73)			.519 (13.18)
22		.922 (23.42)	1.656 (42.06)	
24	2.900 (73.66)	.984 (24.99)	1.750 (44.45)	.657 (16.69)

Shell size	N free dia	ZZ screw threads	YZ lockwashers NASM35338 or NASM35333
8	.125 (3.18)	6-32 UNC	NASM35338-98 or -41 NASM35333-105 or -37
10	.188 (4.78)		
12	.312 (7.92)		
14	.375 (9.53)		
16	.500 (12.70)		
18	.625 (15.88)	8-32 UNC	NASM35338-99 or -42 NASM35333-106 or -38
20	.625 (15.88)		
22	.750 (19.05)		
24	.800 (20.32)		

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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Dimensions for class P

Shell size	L max over-all length	X dia min ID	Y dia max OD	PP min
8	1.656 (42.06)	.317 (8.05)	.608 (15.44)	.250 (6.35)
10		.434 (11.02)	.734 (18.64)	
12		.548 (13.92)	.858 (21.79)	
14		.673 (17.09)	.984 (24.99)	
16		.798 (20.27)	1.110 (28.19)	
18		.899 (22.83)	1.234 (31.34)	
20	1.750 (44.45)	1.024 (26.01)	1.360 (34.54)	
22	1.750 (44.45)	1.149 (29.18)	1.484 (37.69)	
24	1.782 (45.26)	1.274 (32.36)	1.610 (40.89)	

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

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NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.
2. Distance between end of shell and the point at which a gage pin having the same basic diameter as the mating contact and a square face first engages socket contact spring.
3. Packing material shall be In accordance with ASTM-D2000.
4. True position (TP) tolerances in accordance with ASME Y14.5.
5. For class F: Lockwashers may be captivated. Use SAE-AS85049/139 bushing with dimension N if reduced opening is required.

FIGURE 1. Plug, dimensions, classes E, F and P - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MS3126.

Insert arrangement shall be in accordance with MIL-STD-1669.

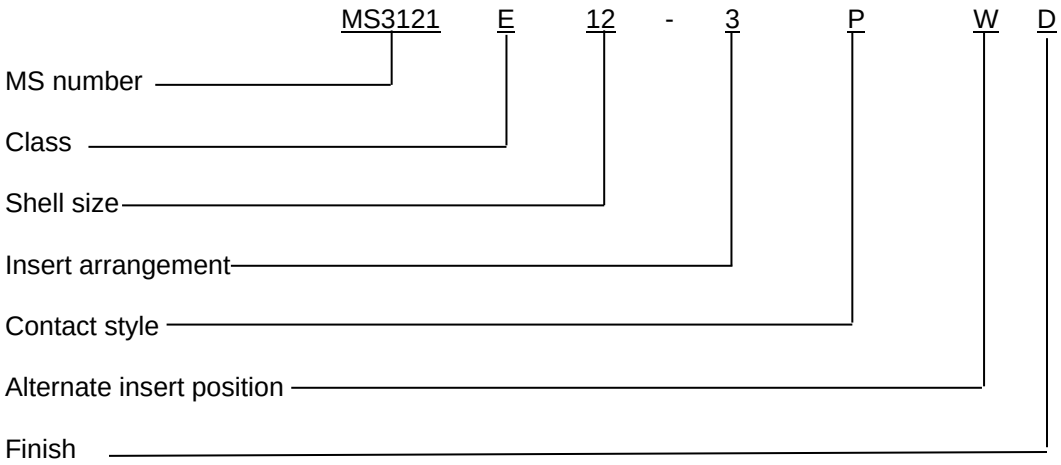
Packing material shall meet the requirements of ASTM-D2000.

Finishes: A finish designator shall not be included in the PIN when finish W is required. See MIL-DTL-26482 for alternative finishes D, T and K for classes E, F and P.

Material:

a. Shell - aluminum alloy.
b. Bayonet pins (B dia) - stainless steel, passivated.

Part or Identifying Number (PIN) example:



Note: No finish designator is used when finish W is required (finish W is the default finish). In the PIN, a dash symbol (“-“) shall be placed between the shell size and insert arrangement for classes E, F and P.

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Referenced documents. In addition to MIL-DTL-26482, this document references the following:

MS3126
MIL-STD-1669
ASME Y14.5
ASTM-D2000
NASM35333
NASM35338
SAE-AS39029/31
SAE-AS39029/32
SAE-AS85049/139

CONCLUDING MATERIAL

Custodians:
Army - CR
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2017-116)

Review activities:
Army - CR4, MI
Air Force - 99

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